

Machine Learning Assignment 2 Submission Report

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1) GitHub Repository Link

Repository URL: https://github.com/2025aa05033/ML_Assignment_2

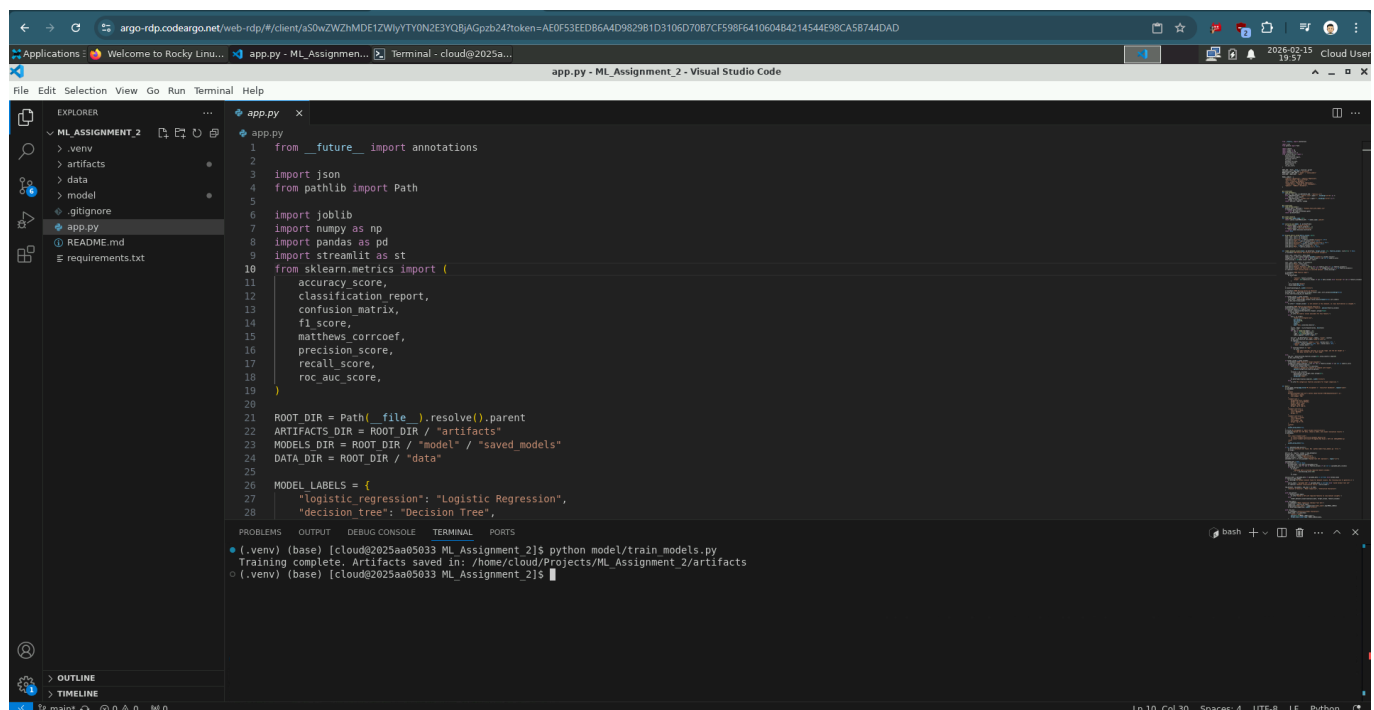
Repository includes:

- Complete source code
- `requirements.txt`
- `README.md`

2) Live Streamlit App Link

App URL: <https://2025aa05033-ml-assignment-2.streamlit.app/>

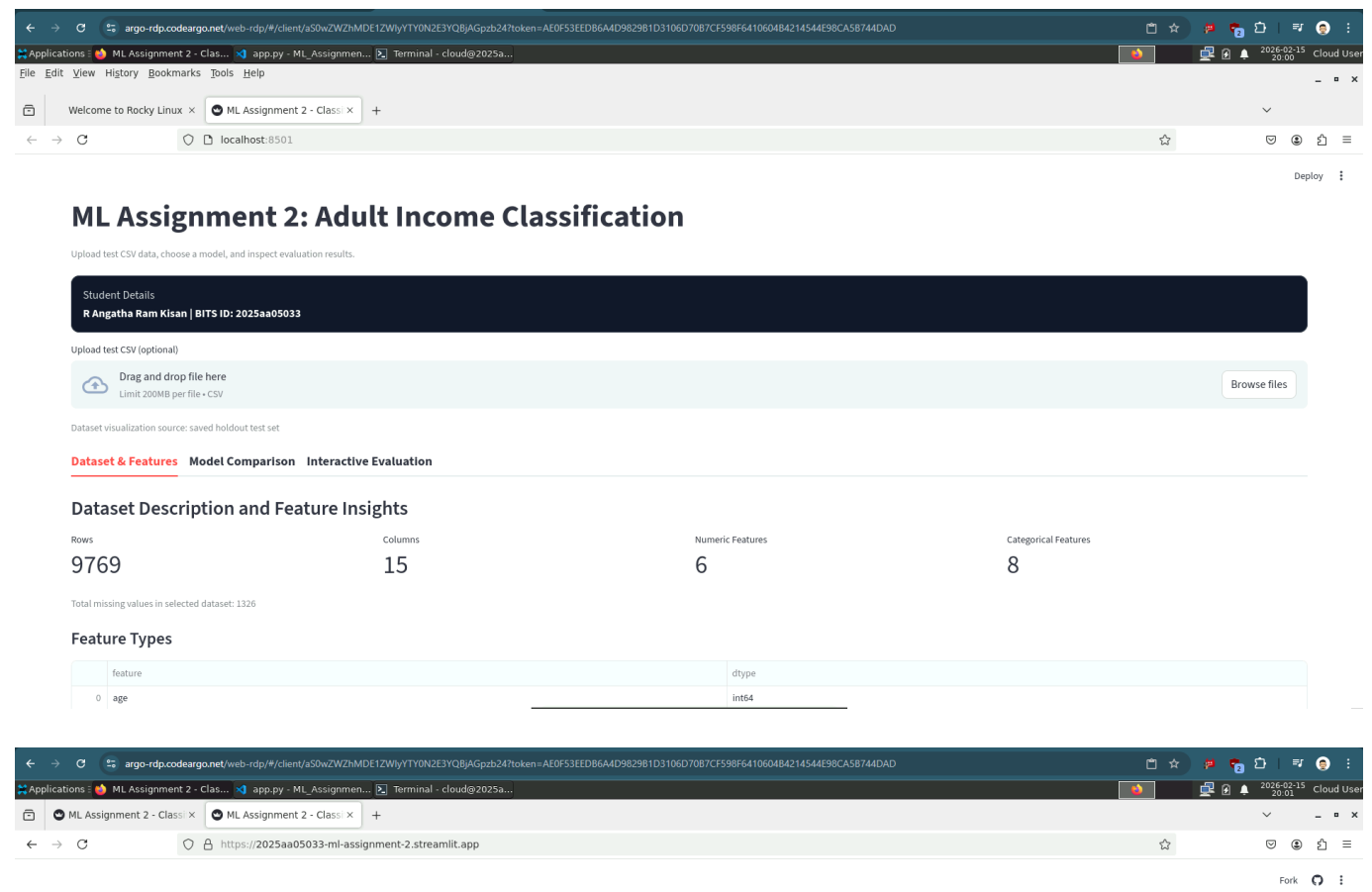
3) Screenshot (BITS Virtual Lab Execution Proof)



```
1 from __future__ import annotations
2
3 import json
4 from pathlib import Path
5
6 import joblib
7 import numpy as np
8 import pandas as pd
9 import streamlit as st
10 from sklearn.metrics import (
11     accuracy_score,
12     classification_report,
13     confusion_matrix,
14     f1_score,
15     matthews_corrcoef,
16     precision_score,
17     recall_score,
18     roc_auc_score,
19 )
20
21 ROOT_DIR = Path(__file__).resolve().parent
22 ARTIFACTS_DIR = ROOT_DIR / "artifacts"
23 MODELS_DIR = ROOT_DIR / "model" / "saved_models"
24 DATA_DIR = ROOT_DIR / "data"
25
26 MODEL_LABELS = {
27     "logistic_regression": "Logistic Regression",
28     "decision_tree": "Decision Tree",
29 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
• (.venv) (base) [cloud@2025aa05033 ML_Assignment_2]$ python model/train_models.py
Training complete. Artifacts saved in: /home/cloud/Projects/ML_Assignment_2/artifacts
• (.venv) (base) [cloud@2025aa05033 ML_Assignment_2]$
```



4) README Content

a) Problem Statement

Build an end-to-end machine learning classification project on one public dataset and implement 6 required models on the same train/test split:

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbor Classifier
4. Naive Bayes Classifier (Gaussian)
5. Random Forest (Ensemble)
6. XGBoost (Ensemble)

Then create an interactive Streamlit app with CSV upload, model selection, metric display, and confusion matrix / classification report.

b) Dataset Description

- **Dataset:** UCI Adult Income dataset (fetched via OpenML: `adult`, version `2`)
- **Task type:** Binary classification (`<=50K` vs `>50K`)
- **Instances:** 48,842
- **Features:** 14 total (mixed numeric + categorical), satisfying assignment minimum of 12 features
- **Target column:** `class` (mapped to `0` for `<=50K`, `1` for `>50K`)
- **Split strategy:** Stratified train/test split (`80/20`, `random_state=42`)

c) Models Used and Metrics

Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8524	0.9042	0.7414	0.5885	0.6562	0.5699
Decision Tree	0.8169	0.7490	0.6171	0.6189	0.6180	0.4976
kNN	0.8384	0.8699	0.6827	0.6065	0.6424	0.5400
Naive Bayes (Gaussian)	0.6204	0.8287	0.3794	0.9213	0.5374	0.3866
Random Forest (Ensemble)	0.8644	0.9163	0.7953	0.5834	0.6731	0.6013
XGBoost (Ensemble)	0.8726	0.9272	0.7943	0.6309	0.7032	0.6301

Observations on Model Performance

ML Model Name	Observation about model performance
Logistic Regression	Strong baseline with balanced performance; lower recall than ensembles but stable and interpretable.
Decision Tree	Captures nonlinearity but overfits relative to ensembles, giving lower AUC and MCC.
kNN	Better than single tree on this dataset, but performance is sensitive to distance behavior in high-dimensional encoded space.
Naive Bayes (Gaussian)	Very high recall but poor precision, producing many false positives and weaker overall balance.
Random Forest (Ensemble)	Better precision and MCC than single-tree models with reduced model size after optimization.
XGBoost (Ensemble)	Best overall model across key ranking metrics, with strong balance after size optimization.